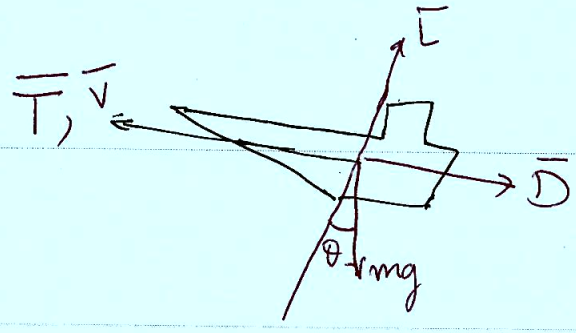
1/22/19 Lecture 3.

Ch2.

Along T, D axis.




$$m\frac{d\bar{V}}{dt} = \bar{T} - \bar{D} - mg\sin\theta$$

Along L axis.

$$\frac{mV^2}{r_c} = mg\cos\theta - L$$

Look @ two different cases.

(a) Horizontal flight $\Rightarrow \theta = 0$

$r_c \to \infty$

& get $L = mg$ (& T axis $\Rightarrow m\frac{dV}{dt} = T - D$)

(b) Satellite

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Remember (low speed aero) $L = \underbrace{q_0}_{\text{dyn pr.}} C_L S$

$\Rightarrow$ no atm $\Rightarrow q_0 = 0$

$\Rightarrow L = 0$

$$\Rightarrow \frac{mV^2}{r_c} = mg \qquad (\theta = 0)$$

$$\Rightarrow V = (r_c g)^{1/2}$$

$g$ ↳ accel. due to gravity.

For LEO alt ~ 330 km

$\Rightarrow r_c = 6700$ km

$V_{LEO} = 8100$ m/s to maintain orbit.

Note @ orbital altitudes Mach # does not make sense b/c pr is very low.

Kn (knudsen #) is not small $\Rightarrow$ not a continuum flow & sound does <u>NOT</u> propagate.

<u>But</u> let's assume $T \sim 200$ K.

$$a = \sqrt{\underset{1.4}{\gamma} RT} = 283 \text{ m/s}$$

$$M = \frac{V_{LEO}}{a} = 28$$

$\Rightarrow$ typically say we want $M \approx 25$ to achieve orbit.

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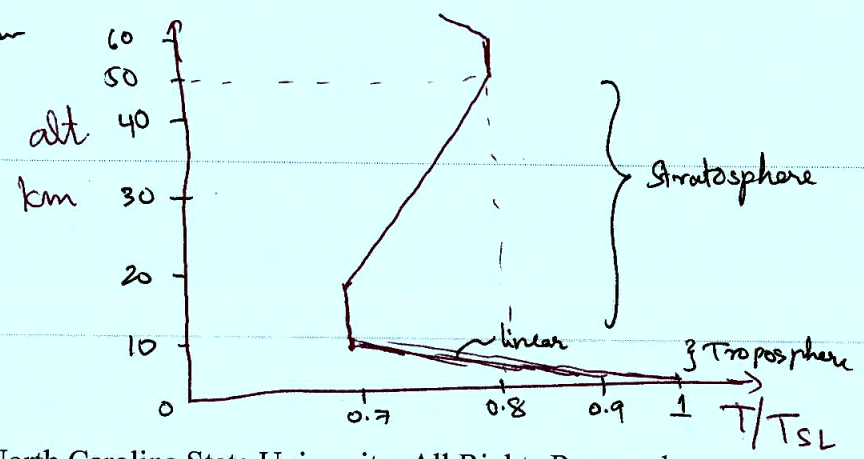
# Standard Atmosphere

- What is the upper limit of the atmosphere?
- It is constantly changing & depends on many factors.
- Most important is how to <u>define</u> the edge?
- The density ($\rho$) decreases exponentially @ high altitudes & approaches zero asymptotically.
- Using a result similar to B.L.T., define the height as the location where 99% of the mass is below this altitude

  $\rightarrow \approx 31$ km.
- The height depends on the solar cycle, which has a 11-yr time period when the thermal radiation incident on the earth repeats itself.

Temperature Profile of Std. Atm




$T_{SL} = 288K$

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- Temp decreases linearly in troposphere (0-11 km)
- Temp is const / increasing linear (very mildly) in stratosphere (11-52 km)
- Typical commercial jets max alt still in troposphere
- Want scramjets to fly in stratosphere due to dyn. pr. considerations.

<u>Transport Properties</u>

- Interested in transport of momentum & energy.
- Mom. transport : assume it scales linearly w/ vel. gradient → Newtonian fluid

$$\tau = \mu \frac{\partial v}{\partial y} \quad \text{for 1D flow}$$

$\mu$ : coeff of viscosity $\left(N \cdot s / m^2\right)$

- Energy transport : scales linearly w/ T gradient.

→ Fourier heat transfer

$$q = -k \frac{dT}{dx} \qquad \text{1-D heat conduction}$$

$k$ : thermal conductivity $\left(\frac{W}{m\,K}\right)$

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→ $\nu = \frac{\mu}{\rho} \left[\frac{m^2}{s}\right]$

$\alpha = \frac{k}{\rho C_p} \left[\frac{m^2}{s}\right]$

Can now compare $q$ & $\tau$ b/c. $\nu$ & $\alpha$ units are the same.

→ $\mu$ (or $\nu$) & $k$ (or $\alpha$) are funct$^n$ of T.

$\nu$ & $\alpha$ scale as $T^{3/2}$ for gases

$\mu$ & $k$ scale as $T^{1/2}$ for gases.

→ For ideal gas, we have empirical formulas.

$$\mu = 1.46e-6 \left[\frac{T^{3/2}}{T+111}\right] \left(\frac{N \cdot s}{m^2}\right) \quad (T \text{ in } K)$$

$$k = 1.99e-3 \left[\frac{T^{3/2}}{T+112}\right] \left(\frac{W}{m \cdot k}\right)$$

Why don't you have a pr dependence?

→ As previously stated, pr decreases exponentially w/ altitude

→ No easy eq$^n$ for $\mu$(alt) or $k$(alt) exists due to complex relation b/t T & alt.

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Recall from 1st Law : derived stag$^n$ relation

$$h_0 = h_\infty + \frac{1}{2} V_\infty^2$$

(*) Note : From this point on, we are going to use '0' for f.s. cond$^n$s. 't' for total cond$^n$.

Hence, $h_t = h_0 + \frac{1}{2} V_0^2$

the above eqn becomes

pr = Force/area.

define dyn. pr. $q_0 = \frac{1}{2} \rho_0 V_0^2$ ; $q_0, \rho_0, V_0 \rightarrow$ freestream values.

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Frictional heating & pressure (forces) will scale w/ $q_0$. So we must pay attn to this parameter for high-speed flows as well.

(Aside: for low-speed aero $L = q_0 C_L S$ ; $D = q_0 C_D S$

In hyp. flows : $p - p_0 = q_0 \underbrace{C_{p_t}}_{\text{stagn pt pr. coeff}} \sin\theta_b$ , where $\theta_b$ is local inclination angle of the body w.r.t. f/s

(weak fn of M) & ranges between 1.75 to 1.8

Since pr in hyp. flow is directly related to $q_0$, all the forces are also directly related to $q_0$ )

Using ideal gas law.

$$q_0 = \frac{1}{2}\rho_0 V_0^2 = \frac{1}{2}\left(\frac{p_0}{RT_0}\right)\cdot(\gamma R T_0) M_0^2$$

$$\boxed{q_0 = \frac{1}{2}\gamma p_0 M_0^2}$$

$p_0$ : local undisturbed static pr.

→ Typically require $20\,kPa \le q_0 \le 90\,kPa$. due to structural & thermal limitations.

→ Thus to fly fast (high M#), we need to fly high (low $p_0$).

Eg: Vehicle $q_0$ must not exceed 60 kPa. What min. altitude must vehicle maintain if $M_0 = 7$?

$$q_0 = \frac{1}{2}\gamma p_0 M_0^2 \Rightarrow p_0 = \frac{2q_0}{\gamma M_0^2} = \frac{2\times 60(kPa)}{1.4\times 7^2} = \cancel{1.75\,kPa}.\ 1.75\,kPa.$$

$$\frac{p_0}{p_{SL}} = \frac{1.75}{101} = 17.5\times10^{-3} \Rightarrow h_{alt} = \underline{27.5\,km}.$$

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